

A SHARP LOCALIZATION CONSTANT FOR GURLAND'S RATIO OF THE GAMMA FUNCTION

DEEPSEEK V4.1 FLASH, GPT 5.6 SOL, GROK 4.7, AND CLAUDE OPUS 5

ABSTRACT. Gurland's ratio $\mathcal{G}(x, y) = \Gamma(x)\Gamma(y)/\Gamma^2(\frac{x+y}{2})$ is the exponential of the Jensen gap of $\log \Gamma$. The gap has the classical expansion

$$\log \mathcal{G}(x, y) = \sum_{k \geq 1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, \frac{x+y}{2} \right),$$

in the form recorded by Srivastava, and the series converges for all $x, y > 0$: its radius is the distance from the midpoint to the pole of Γ at the origin. The same specialization for the modified ratio $\mathcal{G}_\star$ therefore holds on the whole quadrant, and the restriction $Q < 1$ is not required for identifying $\log \mathcal{G}_\star$ with its series. That identification settles two questions posed by Wiśniewska (arXiv:2512.07028); it is the classical expansion at the point $1 + \frac{x+y}{2}$. We give two-sided remainder estimates whose geometric rate $\rho = |x-y|/(2+x+y)$ is optimal in the sense that $\frac{1}{m} \log R_m \rightarrow 2 \log \rho$. The prefactors on the two sides may differ by a large factor. The rate satisfies $\rho \leq Q$, with equality only for $x = y$, while Q is unbounded. For large arguments, $\log \mathcal{G}_\star = (1+v) \Lambda(\rho) (1+o(1))$, where $\Lambda(\rho) = 2(\log 2 - H_b(\frac{1+\rho}{2}))$ and H_b is the binary entropy formed with the same logarithm. This is a leading-order statement. The expansion extends to any number of variables, with applications to the beta function and to certified computation. The principal result localizes the mean-value parameter $t(x, y)$ in the two-sided bound of Yang and Zheng. Writing

$$\theta(x, y) = \frac{t(x, y) - \sqrt{xy}}{\frac{x+y}{2} - \sqrt{xy}},$$

we prove $\theta(x, y) > \frac{2}{3}$. The constant $\frac{2}{3}$ is optimal and is not attained. The proof is self-contained. High-precision checks accompany the statements.

1. INTRODUCTION

1.1. **Background.** For $x, y > 0$ Gurland's ratio [6] is

$$(1) \quad \mathcal{G}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma^2(\frac{x+y}{2})},$$

and Gurland's inequality states that $\mathcal{G}(x, y) > 1$ for $x \neq y$. Since $\log \Gamma$ is strictly convex on $(0, \infty)$ (Bohr–Møllerup), $\log \mathcal{G}(x, y)$ is exactly the Jensen gap

$$(2) \quad \log \mathcal{G}(x, y) = \log \Gamma(x) + \log \Gamma(y) - 2 \log \Gamma\left(\frac{x+y}{2}\right) \geq 0,$$

so Gurland's ratio is the exponential of the second-order Jensen gap of $\log \Gamma$. The modified ratio

$$(3) \quad \mathcal{G}_\star(x, y) = \frac{\Gamma(1+x)\Gamma(1+y)}{\Gamma^2(1+\frac{x+y}{2})} = \frac{4xy}{(x+y)^2} \mathcal{G}(x, y)$$

is the form studied in [17]. The ratio (1) and its relatives were studied by Gautschi [5], Kairies [7], Merkle [10], Alzer [2] and Chen–Choi [4]; see [8] for a survey.

Writing

$$(4) \quad u := \frac{x-y}{2}, \quad v := \frac{x+y}{2}, \quad \text{so that } x = v+u, \quad y = v-u,$$

Date: September 22, 2026.

2020 Mathematics Subject Classification. Primary 33B15; Secondary 11M35, 26D07, 41A58.

Key words and phrases. Gurland's ratio, gamma function, Hurwitz zeta function, Jensen gap, trigamma mean, remainder estimates.

the Weierstrass product yields the finite expansion

$$(5) \quad \log \mathcal{G}_\star(x, y) = \sum_{k=1}^{m-1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta\left(2k, 1 + \frac{x+y}{2}\right) + R_m(x, y), \quad m \geq 2,$$

with the remainder estimate

$$(6) \quad 0 \leq R_m(x, y) \leq \frac{1}{m} \left(\frac{x-y}{2} \right)^{2m} \sum_{n=1}^{\infty} \frac{1}{(n+x)^m (n+y)^m},$$

where $\zeta(s, a) = \sum_{n \geq 0} (n+a)^{-s}$ is the Hurwitz zeta function [17, Thm. 1] (the remainder is denoted R_n^m there). From (5)–(6) it is deduced in [17, Cor. 1] that the condition

$$(7) \quad Q(x, y) := \frac{|x-y|}{2(1+\sqrt{xy})} < 1$$

implies $\log \mathcal{G}_\star(x, y) = S_\infty(x, y)$, where $S_\infty = \lim_{m \rightarrow \infty} S_m$ and S_m denotes the partial sum in (5). The same source gives the bilateral inequality

$$(8) \quad \left(\frac{x-y}{2} \right)^2 \zeta\left(2, 1 + \frac{x+y}{2}\right) \leq \log \mathcal{G}_\star(x, y) \leq \left(\frac{x-y}{2} \right)^2 \zeta(2, 1 + \sqrt{xy}).$$

This two-sided bound is a case of Corollary 11 and Remark 5 of Yang and Zheng [20]. With base point 1, $p = x$ and $q = y$, those results state that

$$(9) \quad \exp \left[\frac{(p-q)^2}{4} \psi'(1+r_1) \right] \leq \frac{\Gamma(1+p)\Gamma(1+q)}{\Gamma^2(1+\frac{p+q}{2})} \leq \exp \left[\frac{(p-q)^2}{4} \psi'(1+r_2) \right],$$

with the best constants $r_1 = \frac{p+q}{2}$ and $r_2 = \sqrt{pq}$ whenever $p, q > 0$. As $\psi' = \zeta(2, \cdot)$ and $\left(\frac{x-y}{2}\right)^2 = u^2$, this is (8). Since $\zeta(2, \cdot)$ is continuous and strictly decreasing, (8) produces a unique parameter $t = t(x, y) \in (\sqrt{xy}, \frac{x+y}{2})$ with

$$(10) \quad \log \mathcal{G}_\star(x, y) = \left(\frac{x-y}{2} \right)^2 \zeta(2, 1+t).$$

Thus the bound (8) and the existence of t are contained in [20]. The question that remains is quantitative: where, inside $(\sqrt{xy}, \frac{x+y}{2})$, the point t lies. The ratio

$$(11) \quad \theta(x, y) = \frac{t(x, y) - \sqrt{xy}}{\frac{x+y}{2} - \sqrt{xy}}$$

measures that position. For $x \neq y$ one has $\theta(x, y) \in (0, 1)$, the endpoint 0 being the geometric mean and the endpoint 1 the arithmetic mean. Theorems 1.8 and 1.9 locate θ .

Three questions are posed in [17, §2].

Open problem 1. Does $R_m(x, y) \rightarrow 0$ hold for every $x, y > 0$?

Open problem 2. Does $\log \mathcal{G}_\star(x, y) = S_\infty(x, y)$ hold in general, and if not, how large can the difference be?

Open problem 3. How is $t(x, y)$ located in the interval $(\sqrt{xy}, \frac{x+y}{2})$? What is its distance to the midpoint of that interval?

1.2. Relation to the literature, and what is new here. The identity in Theorem 1.1 is a symmetrised Taylor expansion and is classical. Since $\psi^{(n)}(z) = (-1)^{n+1} n! \zeta(n+1, z)$ for $\operatorname{Re} z > 0$ and $n \geq 1$ [12, §5.15], the Taylor series of $\log \Gamma$ about $a > 0$ is

$$(12) \quad \log \Gamma(a+w) = \log \Gamma(a) + \psi(a)w + \sum_{n \geq 2} \frac{(-1)^n}{n} \zeta(n, a) w^n, \quad |w| < a.$$

The linear term is $\psi(a)w$. It cannot be written as $-\zeta(1, a)w$, because the Hurwitz zeta function has a pole at $s = 1$. The odd terms cancel in the symmetric difference, and the even part is

exactly (14); equivalently,

$$(13) \quad \log \Gamma(a+w) + \log \Gamma(a-w) - 2 \log \Gamma(a) = \sum_{k \geq 1} \frac{\zeta(2k, a)}{k} w^{2k}, \quad |w| < a,$$

and it is in this form, as a sum of a series of Hurwitz zeta values, that the identity is recorded in the literature (see, e.g., [14]) and in monographs on zeta series [15]; the classical relation between the gamma function and the Hurwitz zeta function goes back at least to [3]. We claim no novelty for (13), nor for its specialization $a = 1 + v$, $w = u$, which is (15). What is new here is (i) two-sided remainder estimates for this series, with geometric rate ρ optimal in the sense $\frac{1}{m} \log R_m \rightarrow 2 \log \rho$, together with the comparison of ρ and Q ; (ii) the observation that the specialization $a = 1 + v$ converges for every $x, y > 0$, which settles Open Problems 1 and 2 of [17] by the classical expansion; (iii) the localization theory of the trigamma mean t : the bracket (23), the expansion (24) as $c \rightarrow 0$, and the bound $\theta > \frac{2}{3}$ of Theorem 1.9, proved by the four-step argument of Section 2; (iv) the leading large-argument asymptotics of Theorem 1.6; (v) the multivariate extension of Theorem 1.7; and (vi) the applications of Section 3.

Gurland's ratio has also been studied intensively by asymptotic methods. Merkle [10] derived bounds and asymptotic expansions; Alzer [1] and Qi [13] gave further bounds and a survey; Tian and Yang [16] established the asymptotic expansion

$$\frac{\Gamma(x+p)\Gamma(x+q)}{\Gamma(x+\frac{p+q}{2})^2} = \exp \left[\sum_{k=1}^n \frac{B_{2k}(\frac{1-|p-q|}{2}) - B_{2k}(\frac{1}{2})}{k(2k-1)(x+\frac{p+q-1}{2})^{2k-1}} + \mathcal{T}_n \right], \quad x \rightarrow \infty,$$

where $p, q \in \mathbb{R}$ with $|p-q| \neq 0$, B_{2k} are the Bernoulli polynomials, and $\mathcal{T}_n$ is the remainder in that expansion. The signed remainder $(-1)^n \mathcal{T}_n$ is completely monotone when $|p-q| < 1$, which yields sharp bounds; the remainder was analysed further by Yang and Tian [18] for the generalized ratio $\frac{\Gamma(x+a)\Gamma(x+b)}{\Gamma(x+c)\Gamma(x+d)}$ with $a+b = c+d$. These expansions are *asymptotic in* $1/x$ with Bernoulli-polynomial coefficients, whereas (14)–(15) are *exact and convergent* in powers of $(\frac{x-y}{2})^2$ with Hurwitz-zeta coefficients. The two descriptions are complementary: the former is adapted to large arguments, the latter to arguments that are close together, and only the latter can be summed to the exact value. Refinements of Gurland's formula for π were given by Lin [9].

Two further strands of the literature are close to the localization problem studied here. Yang and Zheng [20] gave necessary and sufficient conditions for four classes of ratios of gamma functions to be logarithmically completely monotonic; their Corollary 11 and Remark 5 yield the two-sided bound (9), and their Propositions 3 and 4 compare double integrals of $\psi^{(n)}$ over rectangles, which is the same weighted-average picture that underlies (23). In [19] the mean $\psi_n^{-1}(\frac{1}{b-a} \int_a^b \psi_n(x+\tau) d\tau) - x$, generated by a *uniform* average of a polygamma function, is shown to increase from $\min(a, b)$ onto $\frac{a+b}{2}$; the parameter t of (10) is the corresponding mean for a *triangular* average of ψ' (see Remark 1.13), so the two objects are of the same type but not the same, and neither the existence of an interior sharp constant nor its value follows from [19]. Very recently, Moustafa and Mahmoud [11] obtained bounds for Gurland-type ratios in terms of the generalized Nielsen beta function. The remainder theory of (14) and the quantitative localization theory developed below do not appear in these works.

1.3. Results of the present paper. Our starting point is that (2) is a second-order symmetric difference of $\log \Gamma$, so the subject is governed by the Taylor expansion of $\log \Gamma$ about the midpoint v . Since the singularities of $\log \Gamma$ nearest to $v > 0$ are the poles of Γ at $0, -1, -2, \dots$, the Taylor series about v converges on the disc $|z-v| < v$, and $x = v+u$, $y = v-u$ both lie in that disc precisely when $|u| < v$, that is, precisely when $x, y > 0$. The expansion therefore converges on the whole quadrant.

Theorem 1.1 (Exact expansion). *For all $x, y > 0$,*

$$(14) \quad \log \mathcal{G}(x, y) = \sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, \frac{x+y}{2} \right),$$

$$(15) \quad \log \mathcal{G}_\star(x, y) = \sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right).$$

Both series converge absolutely. Moreover (14) converges for exactly those $(x, y) \in \mathbb{R}^2$ with $|x - y| < x + y$, and (15) for exactly those with $|x - y| < 2 + x + y$; on the quadrant $x, y > 0$ both hold identically, and the two regions are optimal.

Corollary 1.2. *For every $x, y > 0$ and every $m \geq 1$,*

$$\begin{aligned} R_m(x, y) &= \log \mathcal{G}_\star(x, y) - \sum_{k=1}^{m-1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right) \\ &= \sum_{k=m}^{\infty} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right) \geq 0. \end{aligned}$$

Consequently $R_m(x, y) \downarrow 0$ and $\log \mathcal{G}_\star(x, y) = S_\infty(x, y)$ for every $x, y > 0$. Open Problems 1 and 2 both have affirmative answers on the whole quadrant.

The next result quantifies the convergence. The relevant rate is

$$(16) \quad \rho(x, y) := \frac{|x-y|}{2+x+y} \quad (\mathcal{G}_\star), \quad \rho_{\mathcal{G}}(x, y) := \frac{|x-y|}{x+y} \quad (\mathcal{G}).$$

Theorem 1.3 (Remainder estimates). *Let $x, y > 0$, $m \geq 1$, $u = \frac{x-y}{2}$, $v = \frac{x+y}{2}$, $\rho = \rho(x, y) = |x-y|/(2+x+y)$ and $R_m := \log \mathcal{G}_\star(x, y) - \sum_{k=1}^{m-1} \frac{u^{2k}}{k} \zeta(2k, 1+v)$. Then*

$$(17) \quad \frac{u^{2m}}{m} \zeta(2m, 1+v) \leq R_m \leq \frac{\rho^{2m}}{m(1-\rho^2)} \left(1 + \frac{1+v}{2m-1} \right).$$

Moreover, if $x \neq y$, then $\lim_{m \rightarrow \infty} \frac{1}{m} \log R_m = 2 \log \rho$, so the geometric rate ρ is optimal. The same holds for $\mathcal{G}$ with ρ replaced by $\rho_{\mathcal{G}} = |x-y|/(x+y)$ and $\zeta(2m, 1+v)$ by $\zeta(2m, v)$.

The limit fixes the base of the exponential. It does not assert that the prefactors in (17) are best possible: the last row of Table 2 has lower bound 0.3147 and upper bound 15.9456.

Proposition 1.4. *For all $x, y > 0$ one has $\rho(x, y) \leq Q(x, y)$, with equality iff $x = y$. Moreover $\rho(x, y) < 1$ for all $x, y > 0$, whereas $\sup_{x, y > 0} Q(x, y) = +\infty$; for instance $Q(x, 1) = \frac{x-1}{2(1+\sqrt{x})} \sim \frac{\sqrt{x}}{2}$ as $x \rightarrow \infty$. Hence $\{Q < 1\}$ is a proper subset of the quadrant on which (15) converges.*

Corollary 1.5. *For all $x, y > 0$ and all $m \geq 1$,*

$$S_m(x, y) \leq \log \mathcal{G}_\star(x, y) \leq S_m(x, y) + \frac{\rho^{2m}}{m(1-\rho^2)} \left(1 + \frac{1+v}{2m-1} \right),$$

where $S_m(x, y) = \sum_{k=1}^{m-1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta(2k, 1+v)$. Taking $m = 2$ improves the upper bound of (8) whenever $\rho < \frac{\sqrt{5}}{3}$, while S_m increases to $\log \mathcal{G}_\star(x, y)$ as $m \rightarrow \infty$.

We also determine the behaviour of the ratio at large arguments. Recall

$$(18) \quad \Lambda(\rho) := (1+\rho) \log(1+\rho) + (1-\rho) \log(1-\rho), \quad |\rho| < 1.$$

One has $\Lambda(\rho) = \sum_{k \geq 1} \rho^{2k} / (k(2k-1))$, $\Lambda(\rho) > 0$ for $\rho \neq 0$, and $\Lambda(\rho) = 2\rho \operatorname{artanh} \rho + \log(1-\rho^2)$. Let H_b denote the binary entropy

$$(19) \quad H_b(p) = -p \log p - (1-p) \log(1-p),$$

formed with the same logarithm as above. For $p = \frac{1+\rho}{2}$,

$$H_b(p) = -\frac{1+\rho}{2} \log \frac{1+\rho}{2} - \frac{1-\rho}{2} \log \frac{1-\rho}{2} = \log 2 - \frac{1}{2} \Lambda(\rho),$$

and therefore

$$(20) \quad \Lambda(\rho) = 2 \left(\log 2 - H_b \left(\frac{1+\rho}{2} \right) \right).$$

As Λ vanishes to second order at the origin it is convenient to divide it by ρ^2 . Set

$$(21) \quad \ell(\rho) := \frac{\Lambda(\rho)}{\rho^2} = \sum_{k \geq 1} \frac{\rho^{2k-2}}{k(2k-1)} \quad (0 < \rho < 1), \quad \ell(0) := 1.$$

The series converges for $|\rho| < 1$, so ℓ is continuous and strictly increasing on $[0, 1)$, with values at least 1, and $\Lambda(\rho) = \rho^2 \ell(\rho)$. The binary entropy enters the asymptotics through (20); the coefficient in Theorem 1.6 is the normalization ℓ .

Theorem 1.6 (Large-argument asymptotics). *Let $x_j, y_j > 0$, $v_j = \frac{x_j+y_j}{2} \rightarrow \infty$, $u_j = \frac{x_j-y_j}{2}$, and put $\rho_j := \frac{|u_j|}{1+v_j}$. Suppose*

$$\rho_j \rightarrow \rho_0 \in [0, 1).$$

Then, with ℓ as in (21),

$$\log \mathcal{G}_*(x_j, y_j) = \frac{u_j^2}{1+v_j} \ell(\rho_j) (1 + o(1)) = (1+v_j) \Lambda(\rho_j) (1 + o(1)) \quad (j \rightarrow \infty),$$

and the analogous statement holds for $\log \mathcal{G}$ with $1+v_j$ replaced by v_j throughout, i.e. under the hypothesis $|u_j|/v_j \rightarrow \rho_0 \in [0, 1)$ one has $\log \mathcal{G} = \frac{u_j^2}{v_j} \ell(\frac{|u_j|}{v_j}) (1 + o(1))$. (The sign of u_j is immaterial, since Λ and ℓ are even.)

The expansion extends verbatim to any number of variables.

Theorem 1.7 (Multivariate expansion). *Let $n \geq 2$, $x_1, \dots, x_n > 0$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. Write $\mu_j = \sum_{i=1}^n (x_i - \bar{x})^j$ for $j \geq 2$. If $\max_i |x_i - \bar{x}| < \bar{x}$, then*

$$(22) \quad \log \frac{\prod_{i=1}^n \Gamma(x_i)}{\Gamma(\bar{x})^n} = \sum_{j=2}^{\infty} \frac{(-1)^j}{j} \zeta(j, \bar{x}) \mu_j,$$

the series converging absolutely. In particular $\prod_{i=1}^n \Gamma(x_i) \geq \Gamma(\bar{x})^n$, with equality iff $x_1 = \dots = x_n$.

Finally we address Open Problem 3. The parameter t is unique, the gap $v - t$ admits an explicit bracket and an expansion in powers of c^2 , and the ratio θ is bounded by the constant $\frac{2}{3}$.

Theorem 1.8 (Localization of t). *Let $x \neq y$ be positive, $v = \frac{x+y}{2}$, $s = \sqrt{xy}$, $c = \frac{|x-y|}{2}$, $A = 1 + v$ and $\rho = c/A$. Then:*

- (i) *there is a unique $t = t(x, y) \in (s, v)$ satisfying (10);*
- (ii) *with $\rho = \rho(x, y)$ one has*

$$(23) \quad \frac{c^2 \zeta(4, A)}{4 \zeta(3, A) + 6v \zeta(4, A)} \leq v - t \leq \frac{c^2 \zeta(4, A)}{4 \zeta(3, A)} \left(1 + \frac{2\rho^2}{3(1-\rho^2)} \right);$$

- (iii) *as $c \rightarrow 0^+$ with v fixed,*

$$(24) \quad v - t = \frac{c^2 \zeta(4, A)}{4 \zeta(3, A)} + \frac{c^4}{2 \zeta(3, A)} \left[\frac{\zeta(6, A)}{3} - \frac{3 \zeta(4, A)^3}{16 \zeta(3, A)^2} \right] + O(c^6).$$

The coefficients in (23) give the scale of $v - t$. They are not optimal: at $(x, y) = (0.1, 199.9)$ one has $v - t = 26.6699$, while the upper bound equals 514.83 (Section 4). The position of t between s and v is settled by the next theorem.

Theorem 1.9 (Sharp localization). *For all $x \neq y > 0$ the ratio (11) satisfies*

$$(25) \quad \theta(x, y) > \frac{2}{3},$$

equivalently $t(x, y) > \frac{1}{3}(x + y + \sqrt{xy})$. The constant $\frac{2}{3}$ is best possible and the bound is not attained: $\theta(x, y)$ tends to $\frac{2}{3}$ along $v \rightarrow \infty$ with $|x - y|/(x + y) \rightarrow 0$.

Remark 1.10 (The midpoint inequality). Since $\frac{2}{3} > \frac{1}{2}$, Theorem 1.9 gives

$$(26) \quad t(x, y) > \frac{1}{2} \left(\sqrt{xy} + \frac{x + y}{2} \right)$$

for every $x \neq y > 0$. The same bound also follows from the upper estimate in (23), on a subset of the quadrant, by the following comparison. That upper bound is strictly smaller than $\frac{v-s}{2} = \frac{c^2}{2(v+s)}$ as soon as

$$\frac{(v + s) \zeta(4, A)}{2 \zeta(3, A)} \left(1 + \frac{2\rho^2}{3(1 - \rho^2)} \right) < 1.$$

The inequality $(n + A)^{-1} \leq A^{-1}$ gives $\zeta(4, A) \leq A^{-1} \zeta(3, A)$, and hence $(v + s) \zeta(4, A) \leq \frac{v+s}{A} \zeta(3, A)$. The display therefore holds whenever

$$(27) \quad \frac{v + s}{2(1 + v)} \left(1 + \frac{2\rho^2}{3(1 - \rho^2)} \right) < 1.$$

A second sufficient condition uses the summands of $\Phi := \log \mathcal{G}_\star(x, y)/c^2$. Let $w = \frac{v-s}{2}$ and $K(\xi) = \frac{1}{c^2} \log \frac{1}{1-c^2/\xi^2}$ for $\xi > c$, and write $M(\xi) = (\xi - w)^{-2}$. Then $\sum_{n \geq 0} K(A + n) = \Phi$ and $\sum_{n \geq 0} M(A + n) = \zeta(2, 1 + \frac{v+s}{2})$, so $t > \frac{v+s}{2}$ if and only if $\sum_{n \geq 0} (K - M)(A + n) < 0$. On $[v + \frac{1}{2}, \infty)$ one has $\xi > c$, and both functions are positive, decreasing and convex: $M''(\xi) = 6(\xi - w)^{-4} > 0$, while $K'(\xi) = -\frac{2}{\xi^3 - c^2\xi}$ and $K''(\xi) = \frac{2(3\xi^2 - c^2)}{(\xi^3 - c^2\xi)^2} > 0$ since $\xi > c$. For a convex decreasing f with $f(\infty) = 0$, the midpoint rule gives $\sum_{n \geq 0} f(A + n) \leq \int_{A-1/2}^{\infty} f$ and the trapezoid rule gives $\sum_{n \geq 0} f(A + n) \geq \int_A^{\infty} f + \frac{f(A)}{2}$. Applying the first to K and the second to M ,

$$\sum_{n \geq 0} (K - M)(A + n) \leq \int_{v+1/2}^{\infty} K - \frac{1}{A - w} - \frac{1}{2(A - w)^2},$$

because $\int_A^{\infty} M = \frac{1}{A-w}$. Consequently $t > \frac{v+s}{2}$ whenever

$$(28) \quad \int_{v+1/2}^{\infty} K - \frac{1}{A - w} - \frac{1}{2(A - w)^2} < 0.$$

The integral equals

$$(29) \quad \int_B^{\infty} K = \frac{1}{c^2} \left[-B \log \frac{B^2}{B^2 - c^2} + c \log \frac{B + c}{B - c} \right], \quad B > c.$$

The proof of Theorem 1.9 is independent of (27) and of (28).

Remark 1.11 (A product formula). The Weierstrass product for Γ gives a representation of $\mathcal{G}_\star$ that makes the identity $\Phi = \sum_{m \geq 0} K(A + m)$ transparent, where $K(\xi) = \frac{1}{c^2} \log \frac{1}{1-c^2/\xi^2}$ and $\Phi = \log \mathcal{G}_\star/c^2$. Indeed $\Gamma(z) = \frac{e^{-\gamma z}}{z} \prod_{n \geq 1} \frac{e^{z/n}}{1 + z/n}$ yields, with $v = \frac{x+y}{2}$ and $c = \frac{|x-y|}{2}$,

$$\frac{\Gamma(v + c)\Gamma(v - c)}{\Gamma(v)^2} = \frac{v^2}{v^2 - c^2} \prod_{n \geq 1} \frac{(n + v)^2}{(n + v)^2 - c^2},$$

the factors $e^{\pm z/n}$ cancelling. Since $\mathcal{G}_\star(x, y) = \frac{4xy}{(x+y)^2} \mathcal{G}(x, y) = \frac{v^2 - c^2}{v^2} \cdot \frac{\Gamma(v+c)\Gamma(v-c)}{\Gamma(v)^2}$, the factors $v^2 - c^2$ cancel as well and

$$(30) \quad \mathcal{G}_\star(x, y) = \prod_{n \geq 1} \frac{(n + v)^2}{(n + v)^2 - c^2} = \prod_{m \geq 0} \frac{(A + m)^2}{(A + m)^2 - c^2},$$

with $A = 1 + v$. Taking logarithms recovers

$$\log \mathcal{G}_\star = \sum_{m \geq 0} \log \frac{(A+m)^2}{(A+m)^2 - c^2} = c^2 \sum_{m \geq 0} K(A+m) = \sum_{k \geq 1} \frac{c^{2k}}{k} \zeta(2k, A),$$

which is Theorem 1.1 for $\mathcal{G}_\star$. Each factor of (30) exceeds 1, because $c < v < A$, and this is the Jensen–Gurland inequality in multiplicative form.

Remark 1.12 (Scaling limit). Fix $\rho = \frac{c}{A}$ and let $A \rightarrow \infty$. Since $\zeta(2m+2, A) \sim \frac{A^{-(2m+1)}}{2m+1}$ and $v - s = A(1 - \sqrt{1 - \rho^2})(1 + o(1))$, the comparison $\zeta(2, A - \Delta) > \Phi$ behind Theorem 1.9 becomes, at leading order in A^{-1} ,

$$(31) \quad \frac{\kappa}{1 - \kappa} > P(\rho^2) - 1, \quad \kappa = \frac{1 - \sqrt{1 - \rho^2}}{3},$$

where P is the series of Step 2 in the proof of Theorem 1.9,

$$(32) \quad P(y) = \sum_{m \geq 0} \frac{y^m}{(m+1)(2m+1)} \quad (0 \leq y < 1),$$

with the closed form $P(\rho^2) = 2 \frac{\operatorname{artanh} \rho}{\rho} + \frac{\log(1 - \rho^2)}{\rho^2}$. The difference of the two sides of (31), $D(\rho) = \frac{\kappa}{1 - \kappa} - P(\rho^2) + 1$, is positive on $(0, 1)$, and this is a consequence of Step 2 of that proof. Indeed, with $y = \rho^2$,

$$\frac{\kappa}{1 - \kappa} = \frac{1 - \sqrt{1 - y}}{2 + \sqrt{1 - y}} = h(y), \quad P(\rho^2) = P(y), \quad \text{so } D = h - P + 1.$$

Step 2 writes $h = \sum_{m \geq 1} a_m y^m$ and $P = 1 + \sum_{m \geq 1} b_m y^m$ with $a_1 = b_1 = \frac{1}{6}$ and $a_m > b_m$ for every $m \geq 2$; hence $D = \sum_{m \geq 2} (a_m - b_m) y^m$ has positive coefficients, which proves $D > 0$ and at the same time identifies the expansion $D(\rho) = \frac{\rho^4}{360} + \frac{11\rho^6}{3024} + O(\rho^8)$, since $a_2 - b_2 = \frac{1}{360}$ and $a_3 - b_3 = \frac{11}{3024}$ are the coefficients computed there (with a positive $O(\rho^8)$ tail). Numerically $D(\rho) \approx \rho^4/360$ down to $\rho = 5 \cdot 10^{-4}$ (where $D = 1.74 \cdot 10^{-16}$) and $D(0.99) = 5.10 \cdot 10^{-2}$. The constant $\frac{2}{3}$ is decided by an $O(\rho^4)$ cancellation, which is why a first-order estimate does not reach Theorem 1.9.

Remark 1.13. Theorem 1.9 can be restated as a comparison of $\zeta(2, \cdot)$ with its own triangular average. With $A = 1 + v$, $c = |x - y|/2$, $K(\xi) = \frac{1}{c^2} \log \frac{1}{1 - c^2/\xi^2}$ and $\Phi = \log \mathcal{G}_\star(x, y)/c^2$, the identity $\sum_{n \geq 0} K(A+n) = \Phi$ of Remark 1.11 combines with $\psi(A+b) - \psi(A-b) = \int_{-b}^b \psi'(A+z) dz$ to give

$$\Phi = \frac{1}{c^2} \int_0^c [\psi(A+b) - \psi(A-b)] db = \int_{-c}^c \psi'(A+z) \frac{c - |z|}{c^2} dz.$$

Since $\zeta(2, 1+t) = \psi'(1+t)$, the inequality of Theorem 1.9 is therefore equivalent to

$$\int_{-c}^c \left[\psi'(A - \Delta) - \psi'(A + z) \right] \frac{c - |z|}{c^2} dz > 0, \quad \Delta = \frac{v - s}{3} = \frac{c^2}{3(v + s)},$$

where $s = \sqrt{xy}$: it compares the value of $\psi' = \zeta(2, \cdot)$ at the point $A - \Delta$ with its average over $[-c, c]$ against the triangular weight $(c - |z|)/c^2$, whose variance is $c^2/6$. Expanding both sides about A , the terms of order c^0 and c^1 cancel (the weight is even), and the first surviving comparison is

$$\frac{\zeta(4, A)}{\zeta(3, A)} < \frac{4}{3(v + s)},$$

which holds for all $x, y > 0$, with relative margin $\frac{1}{A}$ because $\zeta(4, A)/\zeta(3, A) = \frac{2}{3A}(1 + O(A^{-1}))$ and $s \leq v$. The decision rests on the next terms, of relative size ρ^2 , and this is why the constant $\frac{2}{3}$ is sharp and is not attained.

Remark 1.14. The limiting statement in Theorem 1.9 follows from Theorem 1.8(iii). Take $x_j = v_j(1 + \varepsilon_j)$, $y_j = v_j(1 - \varepsilon_j)$ with $v_j \rightarrow \infty$, $\varepsilon_j \rightarrow 0$ and $v_j \varepsilon_j \rightarrow 0$. Then $c_j = v_j \varepsilon_j \rightarrow 0$, so (24) applies and, since $\zeta(4, A)/\zeta(3, A) \sim 2/(3A)$ as $A \rightarrow \infty$,

$$v_j - t_j \sim \frac{v_j^2 \varepsilon_j^2}{4} \cdot \frac{2}{3(1 + v_j)} \sim \frac{v_j \varepsilon_j^2}{6}, \quad v_j - s_j = \frac{v_j \varepsilon_j^2}{1 + \sqrt{1 - \varepsilon_j^2}} \sim \frac{v_j \varepsilon_j^2}{2}.$$

Hence $(v_j - t_j)/(v_j - s_j) \rightarrow \frac{1}{3}$ and $(t_j - s_j)/(v_j - s_j) \rightarrow \frac{2}{3}$. Along this path $\theta = \frac{2}{3} + \frac{1}{6v} + O(v^{-2})$. The column $\varepsilon = v^{-2}$ of Table 4 is such a path and confirms the expansion to nine significant digits. A fixed ε produces a larger limit: $\theta \rightarrow \frac{2}{3} + g(\varepsilon)$ with $g(\varepsilon) > 0$. Numerically $g(\varepsilon) \sim \varepsilon^2/180$ as $\varepsilon \rightarrow 0$; for instance $g(10^{-1}) = 5.60 \cdot 10^{-5}$, while $10^{-2}/180 = 5.56 \cdot 10^{-5}$.

2. PROOFS

2.1. Proof of Theorem 1.1. Let $v > 0$ and put $\Omega = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. The principal branch of $\log \Gamma$ is holomorphic on Ω , and the disc $D(v, v) = \{z : |z - v| < v\}$ is contained in Ω . The nearest singularities of $\log \Gamma$ to v are the poles of Γ at $0, -1, -2, \dots$, all at distance $\geq v$ from v , and $|0 - v| = v$; hence the Taylor series of $\log \Gamma$ about v has radius of convergence exactly v and converges to $\log \Gamma$ on $D(v, v)$.

Fix u with $|u| < v$. Then $v \pm u \in D(v, v)$ and

$$\log \Gamma(v + w) = \log \Gamma(v) + \sum_{n=1}^{\infty} \frac{\psi^{(n-1)}(v)}{n!} w^n, \quad |w| < v,$$

where $\psi = \Gamma'/\Gamma$ and $\psi^{(n)}$ are the polygamma functions. Evaluating at $w = u$ and $w = -u$ and adding,

$$\log \Gamma(v + u) + \log \Gamma(v - u) - 2 \log \Gamma(v) = 2 \sum_{k=1}^{\infty} \frac{\psi^{(2k-1)}(v)}{(2k)!} u^{2k}.$$

The relation $\psi^{(n)}(z) = (-1)^{n+1} n! \zeta(n+1, z)$ for $\operatorname{Re} z > 0$ (see e.g. [12, §5.15]) gives, for $n = 2k - 1$, $\psi^{(2k-1)}(v) = (2k - 1)! \zeta(2k, v)$. Therefore

$$\log \mathcal{G}(x, y) = 2 \sum_{k=1}^{\infty} \frac{(2k - 1)!}{(2k)!} \zeta(2k, v) u^{2k} = \sum_{k=1}^{\infty} \frac{u^{2k}}{k} \zeta(2k, v),$$

which is (14); its region of convergence is $|u| < v$, i.e. $\{|x - y| < x + y\}$, and since all terms are non-negative the convergence is absolute. The same argument applied to $z \mapsto \log \Gamma(1 + z)$ at the point $1 + v$, whose singularities are $\{-1, -2, -3, \dots\}$ and hence lie at distance $\geq 1 + v$ from $1 + v$, with equality at -1 , yields (15) with convergence exactly for $|u| < 1 + v$, i.e. for $|x - y| < 2 + x + y$. Finally, $|x - y| < x + y$ and $|x - y| < 2 + x + y$ both hold for all $x, y > 0$; the first fails for $(x, y) = (1, -\frac{1}{2})$ and the second for $(x, y) = (3, -2)$, so the stated regions are optimal. $\square$

2.2. Proof of Corollary 1.2. By Theorem 1.1, for every $x, y > 0$ the series (15) converges and all its terms are positive. Hence $R_m(x, y) = \sum_{k=m}^{\infty} \frac{u^{2k}}{k} \zeta(2k, 1 + v)$ is the tail of a convergent series of positive terms; it is positive and decreases to 0. This is Open Problems 1 and 2. $\square$

2.3. Proof of Theorem 1.3. Put $A = 1 + v$, $c = |u|$, $\rho = c/A \in (0, 1)$. We use the integral-test estimate

$$(33) \quad \zeta(2k, A) \leq A^{-2k} \left(1 + \frac{A}{2k - 1}\right), \quad k \geq 1, A > 0,$$

obtained from $\zeta(2k, A) = \sum_{n \geq 0} (n + A)^{-2k} \leq A^{-2k} + \int_0^\infty (\tau + A)^{-2k} d\tau$.

Lower bound. All terms of the tail are positive, so $R_m \geq \frac{u^{2m}}{m} \zeta(2m, A)$.

Upper bound. By (33),

$$R_m \leq \sum_{k=m}^{\infty} \frac{c^{2k}}{k} A^{-2k} \left(1 + \frac{A}{2k-1}\right) \leq \left(1 + \frac{A}{2m-1}\right) \sum_{k=m}^{\infty} \frac{\rho^{2k}}{k} = \left(1 + \frac{A}{2m-1}\right) \rho^{2m} \sum_{j=0}^{\infty} \frac{\rho^{2j}}{m+j},$$

and $\sum_{j \geq 0} \frac{\rho^{2j}}{m+j} \leq \frac{1}{m(1-\rho^2)}$, which is (17) for $\mathcal{G}_\star$.

Optimality. Let $x \neq y$, so $c > 0$ and $\rho \in (0, 1)$. Write $\zeta(2m, A) = A^{-2m}(1 + \omega_m)$; since $\sum_{n \geq 1} \left(\frac{A}{n+A}\right)^{2m} \leq \int_0^\infty \left(\frac{A}{A+\tau}\right)^{2m} d\tau = \frac{A}{2m-1}$, we have $0 \leq \omega_m \leq \frac{A}{2m-1} \rightarrow 0$. Hence $R_m \geq \frac{c^{2m}}{m} \zeta(2m, A) = \frac{1+\omega_m}{m} \rho^{2m}$, so $\frac{1}{m} \log R_m \geq 2 \log \rho + o(1)$; conversely (17) gives $\frac{1}{m} \log R_m \leq 2 \log \rho + \frac{1}{m} \log \frac{1+\frac{A}{2m-1}}{m(1-\rho^2)} = 2 \log \rho + o(1)$. The proof for $\mathcal{G}$ is identical with A replaced by v ; there $\zeta(2k, v) \leq v^{-2k} \left(1 + \frac{v}{2k-1}\right)$ and $\rho_{\mathcal{G}} = c/v$. $\square$

2.4. Proof of Proposition 1.4. Since $x+y \geq 2\sqrt{xy}$, $\rho = \frac{|x-y|}{2+x+y} \leq \frac{|x-y|}{2+2\sqrt{xy}} = Q$, with equality iff $x = y$. Also $|x-y| < x+y < 2+x+y$, so $\rho < 1$. Finally $Q(x, 1) = \frac{x-1}{2(1+\sqrt{x})} \sim \frac{\sqrt{x}}{2} \rightarrow \infty$. $\square$

2.5. Proof of Corollary 1.5. The two-sided inequality is Theorem 1.3 restated, and $S_m \rightarrow \log \mathcal{G}_\star(x, y)$ as $m \rightarrow \infty$ by Corollary 1.2. It remains to compare, at $m = 2$, the correction in (17) with the gain over the upper bound of (8). Write $s = \sqrt{xy}$, $c = |u|$ and $A = 1+v$, so that $\rho = c/A$ and

$$c^2 [\zeta(2, 1+s) - \zeta(2, 1+v)] = 2c^2 \int_{1+s}^{1+v} \zeta(3, z) dz \geq 2c^2 \int_{1+s}^{1+v} \frac{dz}{2z^2} = \frac{c^2(v-s)}{(1+s)(1+v)},$$

because $\zeta(3, z) = \sum_{n \geq 0} (n+z)^{-3} \geq \int_0^\infty (\tau+z)^{-3} d\tau = \frac{1}{2z^2}$ (the integrand decreases, so $\int_n^{n+1} (\tau+z)^{-3} d\tau \leq (n+z)^{-3}$ for every $n \geq 0$). Since $v-s = \frac{v^2-s^2}{v+s} = \frac{c^2}{v+s}$ and $(v+s)(1+s) \leq 2v(1+v)$, the gain is at least $\frac{c^4}{2v(1+v)^2}$. The correction in (17) at $m = 2$ is

$$\frac{\rho^4}{2(1-\rho^2)} \left(1 + \frac{A}{3}\right) = \frac{c^4(A+3)}{6A^4(1-\rho^2)},$$

which is smaller as soon as $\frac{v(A+3)}{3A^2(1-\rho^2)} < 1$. As $\frac{v(A+3)}{3A^2} = \frac{v(v+4)}{3(v+1)^2} \leq \frac{4}{9}$, with equality at $v = 2$, this holds whenever $\rho < \frac{\sqrt{5}}{3}$. $\square$

2.6. Proof of Theorem 1.6. If $c_j = 0$, i.e. $x_j = y_j$ for some j , then $\log \mathcal{G}_\star(x_j, y_j) = 0$ and the assertion is trivial, so we may assume $c_j > 0$. Put $A_j = 1 + v_j$, $c_j = |u_j|$, $\rho_j = c_j/A_j \in (0, 1)$, so $\rho_j \rightarrow \rho_0$ and $\Phi_j := \frac{\log \mathcal{G}_\star(x_j, y_j)}{c_j^2} = \sum_{k \geq 1} \frac{c_j^{2k-2}}{k} \zeta(2k, A_j)$. Set $\beta_k(A) := \frac{A^{2k-1}}{k} \zeta(2k, A)$, so that $A_j \Phi_j = \sum_{k \geq 1} \rho_j^{2k-2} \beta_k(A_j)$. Since $\frac{A^{2k-1}}{(n+A)^{2k}} = \frac{1}{A} \left(\frac{A}{n+A}\right)^{2k}$ and $\tau \mapsto \left(\frac{A}{A+\tau}\right)^{2k}$ is decreasing, the integral test gives, for all $k \geq 1$ and $A > 0$,

$$(34) \quad 0 \leq \beta_k(A) = \frac{1}{kA} \sum_{n \geq 0} \left(\frac{A}{n+A}\right)^{2k} \leq \frac{1}{kA} \left(1 + \int_0^\infty \left(\frac{A}{A+\tau}\right)^{2k} d\tau\right) = \frac{1}{kA} + \frac{1}{k(2k-1)}.$$

Hence, for any ρ_1 with $\rho_0 < \rho_1 < 1$ and all large j ,

$$\rho_j^{2k-2} \beta_k(A_j) \leq \rho_1^{2k-2} \left(\frac{1}{k} + \frac{1}{k(2k-1)}\right) \quad (k \geq 1),$$

and the dominating series converges, so dominated convergence applies. On the other hand, for each fixed k the Riemann sum $\frac{1}{A} \sum_{n \geq 0} \left(\frac{A}{n+A}\right)^{2k}$ converges to $\int_0^\infty (1+\tau)^{-2k} d\tau = \frac{1}{2k-1}$ as $A \rightarrow \infty$, so $\beta_k(A) \rightarrow \frac{1}{k(2k-1)}$. Therefore, by the definition (21) of ℓ ,

$$A_j \Phi_j \rightarrow \sum_{k \geq 1} \frac{\rho_0^{2k-2}}{k(2k-1)} = \ell(\rho_0),$$

the case $\rho_0 = 0$ being the definition $\ell(0) = 1$, and the case $\rho_0 > 0$ following from the identity $\sum_{k \geq 1} z^{k-1}/(k(2k-1)) = \Lambda(\sqrt{z})/z$ for $0 < z < 1$, which is obtained by integrating $\sum_{k \geq 1} z^{k-1}/(2k-1) = \operatorname{artanh} \sqrt{z}/\sqrt{z}$ term by term. Since $c_j = A_j \rho_j$, the identity $\log \mathcal{G}_\star = c_j^2 \Phi_j$ reads $\log \mathcal{G}_\star = \rho_j^2 A_j \cdot A_j \Phi_j$; as ℓ is continuous at ρ_0 ,

$$\log \mathcal{G}_\star(x_j, y_j) = \frac{u_j^2}{1 + v_j} \ell(\rho_j)(1 + o(1)) = (1 + v_j) \Lambda(\rho_j)(1 + o(1)),$$

the second equality being $\Lambda(\rho) = \rho^2 \ell(\rho)$. The proof for $\mathcal{G}$ is the same with A_j replaced by v_j and ρ_j by $|u_j|/v_j$, and gives $\log \mathcal{G} = \frac{u_j^2}{v_j} \ell\left(\frac{|u_j|}{v_j}\right)(1 + o(1))$. $\square$

2.7. Proof of Theorem 1.7. Each x_i satisfies $|x_i - \bar{x}| < \bar{x}$, so x_i lies in the disc $D(\bar{x}, \bar{x}) \subset \{\operatorname{Re} z > 0\}$ on which $\log \Gamma$ is holomorphic with radius of convergence exactly $\bar{x}$. Hence

$$\log \Gamma(x_i) = \log \Gamma(\bar{x}) + \sum_{m \geq 1} \frac{\psi^{(m-1)}(\bar{x})}{m!} (x_i - \bar{x})^m$$

converges for every i . Summing over i and using $\sum_i (x_i - \bar{x}) = 0$,

$$\sum_{i=1}^n \log \Gamma(x_i) - n \log \Gamma(\bar{x}) = \sum_{m \geq 2} \frac{\psi^{(m-1)}(\bar{x})}{m!} \mu_m = \sum_{m \geq 2} \frac{(-1)^m}{m} \zeta(m, \bar{x}) \mu_m,$$

which is (22); absolute convergence follows from $|\mu_m| \leq n \max_i |x_i - \bar{x}|^m$ and $|x_i - \bar{x}| < \bar{x}$. The inequality is Jensen's inequality for the convex function $\log \Gamma$, with equality only when all x_i coincide. $\square$

2.8. Proof of Theorem 1.8. (i) *Uniqueness.* The function $g(z) = \zeta(2, 1 + z)$ is continuous and strictly decreasing on $(0, \infty)$. By (8) and the positivity of the terms of (15),

$$\left(\frac{x-y}{2}\right)^2 \zeta(2, 1+v) \leq \log \mathcal{G}_\star(x, y) \leq \left(\frac{x-y}{2}\right)^2 \zeta(2, 1+s),$$

so $\Phi := \log \mathcal{G}_\star(x, y)/c^2$ satisfies $g(v) \leq \Phi \leq g(s)$, both inequalities being strict for $x \neq y$. Hence exactly one $t \in (s, v)$ satisfies $g(t) = \Phi$.

(ii) *The bracket.* Write $\delta = v - t \in (0, v - s)$ and $A = 1 + v$. From (15),

$$(35) \quad \Phi = \zeta(2, A) + \frac{c^2}{2} \zeta(4, A) + W, \quad W := \sum_{k \geq 3} \frac{c^{2k-2}}{k} \zeta(2k, A) \geq 0.$$

Taylor expansion of $z \mapsto \zeta(2, z)$ about A , all of whose terms are positive, gives

$$(36) \quad \zeta(2, A - \delta) = \zeta(2, A) + 2\delta \zeta(3, A) + 3\delta^2 \zeta(4, A) + U, \quad U := \sum_{j \geq 3} (j+1) \delta^j \zeta(j+2, A) \geq 0.$$

Since $\zeta(2, A - \delta) = \Phi$, subtraction yields

$$(37) \quad 2\delta \zeta(3, A) + 3\delta^2 \zeta(4, A) + U = \frac{c^2}{2} \zeta(4, A) + W.$$

For the lower bound write $\zeta_j := \zeta(j, A)$ and put

$$(38) \quad \delta_0 := \frac{c^2 \zeta_4}{4\zeta_3 + 6v\zeta_4}.$$

We prove $\Phi > \zeta(2, A - \delta_0)$, which by strict decrease of $z \mapsto \zeta(2, z)$ gives $\delta > \delta_0$, i.e. the lower bound in (23). By (35) we have $\Phi \geq \zeta_2 + \frac{c^2}{2} \zeta_4$ (as $W \geq 0$), while by (36), $\zeta(2, A - \delta_0) = \zeta_2 + 2\delta_0 \zeta_3 + 3\delta_0^2 \zeta_4 + U_0$ with $U_0 := \sum_{j \geq 3} (j+1) \delta_0^j \zeta_{j+2} \geq 0$. It therefore suffices to verify

$$(39) \quad U_0 < \frac{c^2}{2} \zeta_4 - 2\delta_0 \zeta_3 - 3\delta_0^2 \zeta_4 = 3\delta_0 \zeta_4 (v - \delta_0),$$

the equality following from (38) together with $\delta_0(2\zeta_3 + 3v\zeta_4) = \frac{c^2}{2} \zeta_4$.

Now $\delta_0 < \frac{c^2 \zeta_4}{6v \zeta_4} = \frac{c^2}{6v} < \frac{v}{6}$, so $\delta_0 < \frac{v}{6} < v < A$. Put $\sigma := \delta_0/A \in (0, \frac{1}{6})$. Since $n + A \geq A$ for $n \geq 0$, we have $\zeta_{j+2} \leq A^{-(j-2)} \zeta_4$ for $j \geq 2$, and hence

$$U_0 \leq \zeta_4 A^2 \sum_{j \geq 3} (j+1) \sigma^j = \zeta_4 A^2 \frac{\sigma^3 (4-3\sigma)}{(1-\sigma)^2}.$$

As $\sigma < \frac{1}{6}$ we have $\frac{4-3\sigma}{(1-\sigma)^2} \leq \frac{4}{(5/6)^2} = \frac{144}{25}$, and as $\delta_0 < \frac{v}{6}$ we have $3(v - \delta_0) > \frac{5v}{2}$. Moreover $\delta_0 < \frac{c^2}{6v} < \frac{v}{6}$, so $\delta_0^2 < \frac{v^2}{36}$. Therefore (39) follows from

$$\zeta_4 \frac{\delta_0^2}{A} \cdot \frac{144}{25} \leq \zeta_4 \cdot \frac{5v}{2}, \quad \text{which is implied by} \quad \frac{144}{25} \cdot \frac{v^2}{36A} \leq \frac{5v}{2},$$

and the last inequality reads $\frac{4v^2}{25A} \leq \frac{5v}{2}$, i.e. $\frac{8v}{25} \leq 5(1+v)$, which holds for every $v > 0$. This proves the lower bound.

For the upper bound we estimate W . Since $n + A \geq A$, for $k \geq 2$ and $n \geq 0$ we have $(n + A)^{-2k} \leq A^{-(2k-4)}(n + A)^{-4}$, hence $\zeta(2k, A) \leq A^{-(2k-4)} \zeta(4, A)$; therefore, substituting $k = j + 3$ in the sum,

$$W \leq \zeta(4, A) \sum_{k \geq 3} \frac{c^{2k-2}}{k A^{2k-4}} = \frac{\zeta(4, A) c^4}{A^2} \sum_{j \geq 0} \frac{\rho^{2j}}{j+3} \leq \frac{\zeta(4, A) c^4}{A^2} \cdot \frac{1}{3(1-\rho^2)} = \frac{c^2}{2} \zeta(4, A) \cdot \frac{2\rho^2}{3(1-\rho^2)},$$

where we used $\frac{1}{j+3} \leq \frac{1}{3}$ and $\rho < 1$. Inserting this into (37) and dropping $U \geq 0$ gives $2\delta \zeta(3, A) \leq \frac{c^2}{2} \zeta(4, A) (1 + \frac{2\rho^2}{3(1-\rho^2)})$, the upper bound in (23).

(iii) *Asymptotics.* Write $\zeta_j := \zeta(j, A)$. From $\Phi = \zeta_2 + \frac{c^2}{2} \zeta_4 + \frac{c^4}{3} \zeta_6 + O(c^6)$ and $\zeta(2, A - \delta) = \zeta_2 + 2\delta \zeta_3 + 3\delta^2 \zeta_4 + O(\delta^3)$, equating and inserting $\delta = \alpha c^2 + \beta c^4 + O(c^6)$ gives $2\zeta_3 \alpha = \frac{\zeta_4}{2}$ and $2\zeta_3 \beta + 3\zeta_4 \alpha^2 = \frac{\zeta_6}{3}$, that is $\alpha = \frac{\zeta_4}{4\zeta_3}$ and $\beta = \frac{1}{2\zeta_3} [\frac{\zeta_6}{3} - \frac{3\zeta_4^3}{16\zeta_3^2}]$, which is (24). $\square$

2.9. Proof of Theorem 1.9. Fix $v > 0$, let $0 < c < v$, $s = \sqrt{v^2 - c^2}$, $A = 1 + v$, $\Delta = \frac{v-s}{3}$ and $\Phi(c) = \log \mathcal{G}_\star(x, y)/c^2 = \sum_{k \geq 1} \frac{c^{2k-2}}{k} \zeta(2k, A)$. Since $\zeta(2, 1+t) = \Phi(c)$ and $\zeta(2, \cdot)$ is strictly decreasing,

$$\theta > \frac{2}{3} \iff t > \frac{2v+s}{3} = v - \Delta \iff \zeta(2, A - \Delta) > \Phi(c) \iff \mathcal{R}(c) > 0,$$

where $\mathcal{R}(c) := \zeta(2, A - \Delta(c)) - \Phi(c)$, and $\mathcal{R}(0) = 0$ by continuity. We prove $\mathcal{R}'(c) > 0$ on $(0, v)$. Because $\Delta' = \frac{c}{3s}$ and $\partial_\Delta \zeta(2, A - \Delta) = 2\zeta(3, A - \Delta)$,

$$(40) \quad \mathcal{R}'(c) = \frac{2c}{3s} \zeta(3, A - \Delta) - \frac{2}{c} E(c), \quad E(c) = \sum_{m \geq 1} \frac{m}{m+1} c^{2m} \zeta(2m+2, A),$$

so it suffices to prove

$$(41) \quad E(c) < \frac{c^2}{3s} \zeta(3, A - \Delta).$$

Step 1 (a global midpoint bound for E). Put $K_c(\xi) = \frac{1}{c^2} \log \frac{1}{1-c^2/\xi^2}$ and $\eta_c(\xi) = \frac{1}{\xi^2 - c^2} - K_c(\xi)$. From $K_c(\xi) = \sum_{j \geq 0} \frac{c^{2j}}{j+1} \xi^{-2j-2}$ and $\frac{1}{\xi^2 - c^2} = \sum_{j \geq 0} c^{2j} \xi^{-2j-2}$,

$$\eta_c(\xi) = \sum_{m \geq 1} \frac{m}{m+1} c^{2m} \xi^{-2m-2} > 0, \quad E(c) = \sum_{n \geq 0} \eta_c(A + n).$$

All coefficients being positive, η_c is positive, decreasing and convex on $\xi > c$; the midpoint rule for a convex function, summed over $n \geq 0$, gives with $B = A - \frac{1}{2} > c$

$$(42) \quad E(c) < \int_B^\infty \eta_c(\xi) d\xi,$$

with no restriction on the size of c^2/A . Term-by-term integration with $y = c^2/B^2 \in (0, 1)$ gives

$$(43) \quad \int_B^\infty \eta_c(\xi) d\xi = \frac{c^2}{6B^3} F(y), \quad F(y) = 6 \sum_{m \geq 1} \frac{m}{(m+1)(2m+1)} y^{m-1}.$$

Step 2 (a one-variable comparison). Let $P(y) = \sum_{m \geq 0} \frac{y^m}{(m+1)(2m+1)}$, so $F = 6P'$, and let $h(y) = \frac{1-\sqrt{1-y}}{2+\sqrt{1-y}}$, for which a direct differentiation gives $6h'(y) = \frac{9}{\sqrt{1-y}(2+\sqrt{1-y})^2}$. The identity $h = \frac{3(1-\sqrt{1-y})-y}{3+y}$, together with $1 - \sqrt{1-y} = \sum_{m \geq 1} c_m y^m$, $c_m = \frac{\binom{2m}{m}}{4^m(2m-1)}$, $\frac{c_m}{c_{m-1}} = \frac{2m-3}{2m}$, shows that $h = \sum_{m \geq 1} a_m y^m$ with $a_1 = \frac{1}{6}$ and $a_m = c_m - \frac{1}{3}a_{m-1}$ for $m \geq 2$. Writing $P = 1 + \sum_{m \geq 1} b_m y^m$, $b_m = \frac{1}{(m+1)(2m+1)}$, we have $a_1 = b_1 = \frac{1}{6}$ and, for $r_m = a_m/c_m$, $r_m = 1 - \frac{2m}{3(2m-3)}r_{m-1}$ with $r_2 = \frac{5}{9}$, $r_3 = \frac{17}{27}$; a two-line induction gives $\frac{3(m-1)}{4m} \leq r_m < \frac{3}{4}$ for every $m \geq 2$. Hence $a_2 - b_2 = \frac{1}{360}$, $a_3 - b_3 = \frac{11}{3024}$, and for $m \geq 4$ $a_m > c_m - \frac{1}{4}c_{m-1} = \frac{3(m-2)}{2(2m-3)}c_m$, while $\lambda_m := \frac{3(m-2)c_m}{2(2m-3)b_m}$ satisfies $\lambda_4 = \frac{135}{128} > 1$ and $\frac{\lambda_{m+1}}{\lambda_m} = \frac{(m-1)(m+2)(2m-3)(2m+3)}{2(m-2)(m+1)^2(2m+1)} > 1$ because $2m^3 - 5m^2 + 5m + 22 > 0$. Thus $a_m > b_m$ for all $m \geq 2$, so $h' > P'$ and

$$(44) \quad F(y) < \frac{9}{\sqrt{1-y}(2+\sqrt{1-y})^2}, \quad 0 < y < 1.$$

Step 3 (the finite- v correction only helps). Let $z = A - \Delta$, $e = \frac{1}{2B} \in (0, 1)$ and $q = \frac{s}{B}$, so that $q = \sqrt{(1-e)^2 - y}$ and, exactly, $z = \frac{B(2+q+4e)}{3}$. A direct computation gives

$$\frac{B^3(z+1)}{sz^3} = L_e(y) := \frac{9(2+q+10e)}{q(2+q+4e)^3},$$

and $L_0(y)$ is precisely the right-hand side of (44). Since $q'(e) = -\frac{1-e}{q}$,

$$\frac{d}{de} \log L_e = -\frac{N(e, q)}{q^2(2+q+4e)(2+q+10e)},$$

where $N(e, q) = 40e^3 + 40e^2q - 12e^2 + 83eq^2 - 32eq - 24e + 2q^3 + q^2 - 8q - 4$. For fixed $e \in (0, 1)$ the function $q \mapsto N(e, q)$ is strictly convex ($\partial_q^2 N = 166e + 12q + 2 > 0$), and on $0 < q < 1 - e$ its endpoint values are $N(e, 0) = 4(e-1)(2e+1)(5e+1) < 0$ and $N(e, 1-e) = 9(e-1)(9e^2 - 2e + 1) < 0$. A convex function lies below the maximum of its endpoint values, so $N(e, q) < 0$ on $0 < q < 1 - e$; hence $e \mapsto L_e(y)$ is strictly increasing and

$$(45) \quad \frac{9}{\sqrt{1-y}(2+\sqrt{1-y})^2} < L_e(y) = \frac{B^3(z+1)}{sz^3}.$$

Step 4 (conclusion). Combining (42)–(45),

$$E(c) < \frac{c^2}{6B^3} F(y) < \frac{c^2}{6B^3} \cdot \frac{B^3(z+1)}{sz^3} = \frac{c^2(z+1)}{6sz^3}.$$

Finally, the trapezoid rule applied to the convex decreasing function $\tau \mapsto (z + \tau)^{-3}$ gives $\zeta(3, z) \geq \int_0^\infty (z + \tau)^{-3} d\tau + \frac{1}{2z^3} = \frac{z+1}{2z^3}$, so $E(c) < \frac{c^2}{3s} \zeta(3, z)$, which is (41). Therefore $\mathcal{R}'(c) > 0$ on $(0, v)$; with $\mathcal{R}(0) = 0$ this gives $\mathcal{R}(c) > 0$, and by (40) and the first display $\theta > \frac{2}{3}$, as claimed. $\square$

3. APPLICATIONS

3.1. Sharpened Gurland inequalities. Since $\log \mathcal{G}(x, y) = \sum_{k \geq 1} \frac{u^{2k}}{k} \zeta(2k, v)$ with all terms positive, and since $\zeta(2k, v) \leq v^{-2k} (1 + \frac{v}{2k-1})$ by the integral test, we obtain a two-sided refinement of Gurland's inequality [6].

Corollary 3.1. *For all $x \neq y > 0$, with $v = \frac{x+y}{2}$ and $\rho_{\mathcal{G}} = \frac{|x-y|}{x+y}$,*

$$\frac{(x-y)^2}{4} \zeta\left(2, \frac{x+y}{2}\right) \leq \log \mathcal{G}(x, y) \leq -\log(1 - \rho_{\mathcal{G}}^2) + v \Lambda(\rho_{\mathcal{G}}),$$

and $\mathcal{G}(x, y) > \exp\left(\frac{(x-y)^2}{4}\zeta\left(2, \frac{x+y}{2}\right)\right) > 1$. Moreover $\log \mathcal{G}(x, y) > \frac{(x-y)^2}{2(x+y)}$.

Proof. The lower bound is the first term of (14). For the upper bound, the terms of $\zeta(2k, v) = \sum_{n \geq 0} (n+v)^{-2k}$ with $n \geq 1$ are dominated by the integral over $[0, \infty)$: since $(\tau+v)^{-2k}$ decreases, $\int_{n-1}^n (\tau+v)^{-2k} d\tau > (n+v)^{-2k}$ for every $n \geq 1$, so $\zeta(2k, v) \leq v^{-2k} + \int_0^\infty (\tau+v)^{-2k} d\tau = v^{-2k}\left(1 + \frac{v}{2k-1}\right)$; then, with $\rho = \rho_{\mathcal{G}}$,

$$\log \mathcal{G}(x, y) \leq \sum_{k \geq 1} \frac{u^{2k}}{k} \left(\frac{1}{v^{2k}} + \frac{v^{1-2k}}{2k-1} \right) = \sum_{k \geq 1} \frac{\rho^{2k}}{k} + v \sum_{k \geq 1} \frac{\rho^{2k}}{k(2k-1)} = -\log(1-\rho^2) + v \Lambda(\rho).$$

The last claim uses $\zeta(2, v) = \sum_{n \geq 0} (n+v)^{-2} > \int_0^\infty (\tau+v)^{-2} d\tau = \frac{1}{v}$, because $\int_n^{n+1} (\tau+v)^{-2} d\tau < (n+v)^{-2}$ for every $n \geq 0$. $\square$

3.2. Bounds for the beta function. Recall $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$ and the duplication formula $\Gamma(2v) = \frac{2^{2v-1}}{\sqrt{\pi}}\Gamma(v)\Gamma(v + \frac{1}{2})$, where $v = \frac{x+y}{2}$. Hence $B(x, y) = \mathcal{G}(x, y) \frac{\sqrt{\pi}\Gamma(v)}{2^{2v-1}\Gamma(v + \frac{1}{2})}$. Combining Corollary 3.1 with Gautschi's inequality $v^{1/2} < \frac{\Gamma(v+1)}{\Gamma(v+1/2)} < (v+1)^{1/2}$ [5] yields the following.

Corollary 3.2. *For all $x, y > 0$, with $v = \frac{x+y}{2}$ and $\rho_{\mathcal{G}} = \frac{|x-y|}{x+y}$,*

$$\frac{\sqrt{\pi} v^{-1/2}}{2^{2v-1}} e^{\frac{(x-y)^2}{4}\zeta(2,v)} < B(x, y) < \frac{\sqrt{\pi} (v+1)^{1/2}}{2^{2v-1} v} e^{v\Lambda(\rho_{\mathcal{G}}) - \log(1-\rho_{\mathcal{G}}^2)}.$$

Proof. Gautschi's inequality at the point v , with exponent $\frac{1}{2}$, reads $v^{1/2} < \frac{\Gamma(v+1)}{\Gamma(v+1/2)} < (v+1)^{1/2}$; since $\Gamma(v+1) = v\Gamma(v)$, this gives $v^{-1/2} < \frac{\Gamma(v)}{\Gamma(v+1/2)} < \frac{(v+1)^{1/2}}{v}$. Insert this into the expression for $B(x, y)$ and use Corollary 3.1. $\square$

Example 3.3. For $(x, y) = (1, 2)$ one has $v = \frac{3}{2}$, $\rho_{\mathcal{G}} = \frac{1}{3}$, $\mathcal{G}(1, 2) = 1.273240\dots$ and $B(1, 2) = \frac{1}{2}$. The bounds of Corollary 3.2 read $0.457050 < 0.5 < 0.622777$.

3.3. Certified computation. The two-sided estimate (17) turns the expansion into an algorithm with a certified error.

Corollary 3.4. *Let $x \neq y > 0$, $\rho = \rho(x, y) \in (0, 1)$, $A = 1 + v$ and $0 < \varepsilon < \frac{1}{2}$. If*

$$m \geq \left\lceil \frac{\log((1+A)/(\varepsilon(1-\rho^2)))}{2\log(1/\rho)} \right\rceil,$$

then $|\log \mathcal{G}_(x, y) - S_m(x, y)| \leq \varepsilon$. Moreover, for fixed x, y the number of terms required does not exceed*

$$\frac{\log(1/\varepsilon) + \log(2+v) + \frac{1}{1-\rho}}{2(1-\rho)},$$

so it is $O\left(\frac{\log(1/\varepsilon)}{1-\rho}\right)$ as $\varepsilon \rightarrow 0$, the implied constant being absolute; the arguments enter only through ρ and through the additive term $\log(2+v)$, which is independent of ε .

Proof. For every $m \geq 1$ we have $2m-1 \geq 1$, hence $1 + \frac{A}{2m-1} \leq 1 + A$ and $\frac{1}{m} \leq 1$; therefore (17) gives $0 \leq R_m \leq \frac{(1+A)\rho^{2m}}{1-\rho^2}$, which is at most ε whenever $\rho^{2m} \leq \frac{\varepsilon(1-\rho^2)}{1+A}$. Taking logarithms gives the stated threshold. For the estimate, use $\log \frac{1}{1-\rho^2} \leq \frac{1}{1-\rho}$ (valid for $0 \leq \rho < 1$, since $\log \frac{1}{1-\rho^2} \leq \frac{\rho^2}{1-\rho^2} \leq \frac{1}{1-\rho}$) and $-\log \rho \geq 1 - \rho$; then

$$m \leq \frac{\log(1+A) + \log \frac{1}{\varepsilon} + \log \frac{1}{1-\rho^2}}{2\log \frac{1}{\rho}} \leq \frac{\log(1/\varepsilon) + \log(2+v) + \frac{1}{1-\rho}}{2(1-\rho)}.$$

The leading term as $\varepsilon \rightarrow 0$ is $\frac{\log(1/\varepsilon)}{2(1-\rho)}$, with the absolute constant $\frac{1}{2}$. $\square$

3.4. The multivariate Jensen–Gurland inequality. Theorem 1.7 with $n = 2$ reproduces Theorem 1.1. For a concrete three-point case take $0 < \varepsilon < \frac{1}{2}$ and

$$x_1 = x_2 = v(1 + \varepsilon), \quad x_3 = v(1 - 2\varepsilon),$$

so that $\bar{x} = v$ and $\mu_j = v^j \varepsilon^j (2 + (-2)^j)$; (22) becomes

$$\log \frac{\Gamma(v(1 + \varepsilon))^2 \Gamma(v(1 - 2\varepsilon))}{\Gamma(v)^3} = 3v^2 \varepsilon^2 \zeta(2, v) + 2v^3 \varepsilon^3 \zeta(3, v) + \frac{9}{2} v^4 \varepsilon^4 \zeta(4, v) + O(\varepsilon^5),$$

all displayed coefficients being positive, in accordance with the multivariate Jensen–Gurland inequality of Theorem 1.7.

4. NUMERICAL VERIFICATION

All computations use `mpmath` at 35–60 decimal digits. The script `verify.py` reports `ok` or `FAIL` on each of its 46 checks. Table 1 compares the exact expansion (15) with $\log \mathcal{G}_\star$ computed from $\log \Gamma$. Table 2 checks the two-sided estimate (17), including the extreme case $(0.1, 199.9)$, where the two sides differ by a factor of about fifty. Table 3 checks the bracket (23); at $(0.1, 199.9)$ the upper bound equals 514.83, against $\delta = 26.6699$. Table 4 traces the approach of the ratio (25) to $\frac{2}{3}$, which is reached only in the joint limit $v \rightarrow \infty$, $\varepsilon \rightarrow 0$, $v\varepsilon \rightarrow 0$.

TABLE 1. Exact expansion: $|\log \mathcal{G}_\star - S_m|$ for the truncated series (15), and the number m of terms used.

x	y	$ \log \mathcal{G}_\star - S_m $	m
1	2	$3.5 \cdot 10^{-51}$	34
1	10	$5.1 \cdot 10^{-48}$	142
0.5	3	$1.0 \cdot 10^{-49}$	68
0.2	0.9	$2.1 \cdot 10^{-51}$	37
3	3.0001	$1.0 \cdot 10^{-61}$	6

TABLE 2. The two-sided estimate (17): $\text{lo} = \frac{u^{2m}}{m} \zeta(2m, 1 + v)$ and hi is the right-hand side of (17). The rate ρ is optimal; the ratio hi/lo is not close to 1 in the last row.

(x, y)	ρ	m	R_m	lo	hi
(1, 10)	0.6923	2	$4.1203 \cdot 10^{-1}$	$3.1211 \cdot 10^{-1}$	$6.9851 \cdot 10^{-1}$
(1, 10)	0.6923	12	$2.2659 \cdot 10^{-5}$	$1.2662 \cdot 10^{-5}$	$3.0165 \cdot 10^{-5}$
(0.5, 3)	0.4545	2	$3.6843 \cdot 10^{-2}$	$3.2677 \cdot 10^{-2}$	$5.1563 \cdot 10^{-2}$
(0.5, 3)	0.4545	12	$6.2365 \cdot 10^{-10}$	$5.0458 \cdot 10^{-10}$	$7.1160 \cdot 10^{-10}$
(0.1, 199.9)	0.9891	12	2.9021	0.3147	15.9456

Several further checks are worth recording. First, at $(x, y) = (100, 1)$ one has $Q(100, 1) = 4.5 > 1$ and $\rho(100, 1) = 0.96117 < 1$, and the expansion (15) converges, with tails $R_{20} = 1.478 \cdot 10^{-1}$, $R_{100} = 4.359 \cdot 10^{-5}$ and $R_{1000} = 5.117 \cdot 10^{-37}$. Second, the rate in Theorem 1.3 is confirmed: for $(x, y) = (2, 7)$ one has $2 \log \rho = -1.5769147$, while the computed values of $\frac{1}{m} \log R_m$ are -1.8184 ($m = 5$), -1.7829 ($m = 10$), -1.7157 ($m = 20$), -1.6635 ($m = 40$), -1.6288 ($m = 80$) and -1.6072 ($m = 160$), approaching the predicted limit monotonically. Third, the three-term expansion (24) is confirmed: for $(x, y) = (10, 11)$ one has $v - t = 0.00378192138106$, while the two-term formula gives 0.00378011106367 and the three-term formula 0.00378191965719 , the residual being $-1.7 \cdot 10^{-9}$. Fourth, the large-argument asymptotics of Theorem 1.6: for $(x, y) = (10^6, 2 \cdot 10^6)$, where $v = 1.5 \cdot 10^6$ and $\rho = \frac{|x-y|}{2+x+y} = 0.333333\dots$, one has $\log \mathcal{G}_\star = 169898.977904\dots$ against the prediction $(1+v)\Lambda(\rho) = \frac{u^2}{1+v}\ell(\rho) = 169898.919012\dots$, a relative error of $3.5 \cdot 10^{-7}$, which decreases as the arguments grow ($2.7 \cdot 10^{-5}$ at $(10^4, 3 \cdot 10^4)$). Fifth,

TABLE 3. The bracket (23) of Theorem 1.8: $\delta = v - t$ against its lower bound lo and the two-term value pre $= \frac{c^2 \zeta(4, A)}{4 \zeta(3, A)}$. The upper bound is pre $\cdot (1 + \frac{2\rho^2}{3(1-\rho^2)})$; at (0.1, 199.9) it equals 514.83.

(x, y)	ρ	$\delta = v - t$	lo	pre
(1, 2)	0.2000	0.0199947	0.0115425	0.0197486
(0.5, 3)	0.4545	0.1186430	0.0635181	0.1108211
(2, 7)	0.4545	0.2194361	0.1090893	0.2063217
(1, 10)	0.6923	0.6573045	0.2924374	0.5586888
(20, 100)	0.6557	4.9818888	2.2129255	4.4074125
(0.1, 199.9)	0.9891	26.6699099	8.2958322	16.5501874
(100, 101)	0.0049	0.0004125	0.0002068	0.0004125

TABLE 4. The ratio $\theta = (t - \sqrt{xy}) / (\frac{x+y}{2} - \sqrt{xy})$ along $x = v(1+\varepsilon)$, $y = v(1-\varepsilon)$. For a fixed ε it decreases to $\frac{2}{3} + g(\varepsilon)$ with $g(\varepsilon) > 0$. Numerically $g(\varepsilon) \sim \varepsilon^2/180$ as $\varepsilon \rightarrow 0$ (for instance $g(10^{-1}) = 5.60 \cdot 10^{-5}$, while $10^{-2}/180 = 5.56 \cdot 10^{-5}$), so the value $\frac{2}{3}$ is approached only in the joint limit $v \rightarrow \infty$, $\varepsilon \rightarrow 0$, $v\varepsilon \rightarrow 0$. The column $\varepsilon = v^{-2}$ is such a path, and there $\theta = \frac{2}{3} + \frac{1}{6v} + O(v^{-2})$, as Remark 1.14 shows.

v	$\varepsilon = 10^{-1}$	$\varepsilon = 10^{-3}$	$\varepsilon = v^{-2}$	limit
10^2	0.668396737	0.668333257	0.668333250	0.666666667
10^3	0.666890050	0.666833339	0.666833333	0.666666667
10^4	0.666739371	0.666683339	0.666683333	0.666666667
10^5	0.666724303	0.666668339	0.666668333	0.666666667
10^6	0.666722796	0.666666839	0.666666833	0.666666667
10^7	0.666722645	0.666666689	0.666666683	0.666666667

Corollaries 3.1 and 3.2 were verified on the sampled configurations; for instance $B(1, 2) = \frac{1}{2}$ lies between 0.457050 and 0.622777, and the three-point expansion displayed in Section 3 agrees with $\log \Gamma$ to 19 digits.

5. CONCLUDING REMARKS

Theorem 1.1 is the classical Hurwitz-zeta expansion (13) of $\log \Gamma$, specialized to the second-order symmetric difference that defines $\log \mathcal{G}_*$. The poles of Γ at the non-positive integers determine both the coefficients and the region of convergence, which on the positive quadrant is the whole domain, so $\log \mathcal{G}_*$ coincides with its series for every $x, y > 0$. The condition $Q < 1$ is not required for that identification, and this settles Open Problems 1 and 2 of [17]. The localization of the parameter t in (10) is the content of Theorems 1.8 and 1.9: uniqueness, the expansion of $v - t$ through order c^4 , the bracket (23), and the bound $\theta > \frac{2}{3}$, whose constant is optimal and is not attained. The midpoint inequality (26) is the case $\theta > \frac{1}{2}$ of that bound. Two directions seem worth pursuing:

- (a) the q -analogue, replacing Γ by the q -gamma function Γ_q , for which the role of $\zeta(2k, \cdot)$ is played by the Jackson q -zeta values, and the sign-regularity of the corresponding kernel;
- (b) quantitative versions of Theorem 1.7, such as remainder estimates uniform in n and a central limit theorem for the corresponding Jensen gaps.

Statements and declarations. The authors have no competing interests to declare. All numerical results reported in Section 4 were produced by the accompanying script `verify.py` and can be reproduced with `mpmath` by running `python verify.py`.

REFERENCES

- [1] H. Alzer, *On some inequalities for the gamma and psi functions*, Math. Comp. **66** (1997), 373–389.
- [2] H. Alzer, *Sharp inequalities for the beta function*, Indag. Math. (N.S.) **12** (2001), 15–21.
- [3] B. C. Berndt, *The gamma function and the Hurwitz zeta-function*, Amer. Math. Monthly **92** (1985), no. 2, 126–130, DOI 10.1080/00029890.1985.11971552.
- [4] C.-P. Chen and J. Choi, *Completely monotonic functions related to Gurland’s ratio for the gamma function*, Math. Inequal. Appl. **20** (2017), 651–659.
- [5] W. Gautschi, *Some mean value inequalities for the gamma function*, SIAM J. Math. Anal. **5** (1974), 282–292.
- [6] J. Gurland, *An inequality satisfied by the gamma function*, Skand. Aktuarietidskr. (1956), 171–172.
- [7] H. Kairies, *An inequality for Krull solutions of a certain difference equation*, in: General Inequalities 3 (Oberwolfach 1981), Internat. Schriftenreihe Numer. Math. **64**, Birkhäuser, Basel, 1983, 277–280.
- [8] A. Laforgia and P. Natalini, *On some inequalities for the gamma function*, Adv. Dyn. Syst. Appl. **8** (2013), 261–267.
- [9] L. Lin, *Further refinements of Gurland’s formula for π* , J. Inequal. Appl. **2013** (2013), 2013:48.
- [10] M. Merkle, *Gurland’s ratio for the gamma function*, Comput. Math. Appl. **49** (2005), 389–406.
- [11] H. Moustafa and M. Mahmoud, *New bounds for Gurland-type ratios in terms of the generalized Nielsen’s beta function*, Gulf J. Math. **23** (2026), no. 2, 1–18, DOI 10.56947/x7dwnj62.
- [12] NIST Digital Library of Mathematical Functions, Release 1.2.4, <https://dlmf.nist.gov/>, §5.15 (polygamma functions), §25.11 (Hurwitz zeta function).
- [13] F. Qi, *Bounds for the ratio of two gamma functions*, J. Inequal. Appl. **2010** (2010), Article ID 493058.
- [14] H. M. Srivastava, *Sums of certain series of the Riemann zeta function*, J. Math. Anal. Appl. **134** (1988), no. 1, 129–140, DOI 10.1016/0022-247X(88)90013-3.
- [15] H. M. Srivastava and J. Choi, *Series Associated with the Zeta and Related Functions*, Kluwer Academic Publishers, Dordrecht, 2001.
- [16] J.-F. Tian and Z.-H. Yang, *Asymptotic expansions of Gurland’s ratio and sharp bounds for their remainders*, J. Math. Anal. Appl. **493** (2021), 124545.
- [17] H. Wiśniewska, *Some inequalities for Gurland’s ratio of the gamma functions*, arXiv:2512.07028 [math.CA], 2025.
- [18] Z.-H. Yang and J.-F. Tian, *Complete monotonicity of the remainder of an asymptotic expansion of the generalized Gurland’s ratio*, Math. Inequal. Appl. **27** (2024), DOI 10.7153/mia-2024-27-18.
- [19] Z.-H. Yang and S.-Z. Zheng, *Monotonicity of a mean related to polygamma functions with an application*, J. Inequal. Appl. **2016** (2016), Art. 216, DOI 10.1186/s13660-016-1155-4.
- [20] Z.-H. Yang and S.-Z. Zheng, *Complete monotonicity and inequalities involving Gurland’s ratios of gamma functions*, Math. Inequal. Appl. **22** (2019), no. 1, 97–109, DOI 10.7153/mia-2019-22-07.